

SID DIAMOND

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Personal statement

Recently completed Master's in Physics at Imperial College London researching directed acyclic graphs (DAGs) within the Theoretical Physics Group, now researching agentic foundations and AI safety. Supplemented Master's thesis with courses in information theory and general relativity, alongside research into how benchmark datasets in supervised machine learning can create bias in hate speech detection systems. My bachelor's dissertation was on uncertainty quantification in transformer algorithms using Fisher information, asymptotic normality conditions and Monte Carlo dropout. Last summer I undertook an internship as lead developer for a minimum viable product multi-agent planning application at a software company. The previous summer I conducted an independent research project under the guidance of the Imperial High Performance Computing (HPC) research group, using transformer algorithms to estimate a per-document probability of actionable insight for climate policy documents.

Education

Imperial College London: Integrated MSci in Physics 2022- Present

- Year One: 2:1 | Year Two: 2:1 | Year Three: 2:1 | Masters Year: On track for 1st class
- First-class results in all programming. Object-Oriented Programming (98.7%)
- Master's thesis on modelling innovation networks and applying citation network analysis to the artificial networks (1st class for literature review and viva. Awaiting report grade).
- Statistics (72.9%)
- Problem solving (79.3%)
- BSc Thesis in Uncertainty in Large Language models (69.50%)
- Communicating Physics (84.1%)
- Currently also studying Information theory, a Sociology essay elective (on benchmark datasets to train AI systems) and General Relativity

Stoke Newington School and Sixth Form 2014 -2021

- A-level Subjects: Further Maths (A*), Maths (A*), Physics (A*), Chemistry (A*), EPQ (A*)
- Chemistry Olympiad Silver 2021
- Kelvin Prize winner 2020 (*Peterhouse College Cambridge University Science Competition*)
- GCSE Subjects: 3 grade 9s, 3 grade 8s, 3 grade 7s

Research Projects

Agentic Foundations AI Research | Dovetail Research Spring 2026

- Conducted paid freelance research for Dovetail Research, an ARIA-funded nonprofit focused on agentic foundations.
- Investigated possible extensions of the Touchette–Lloyd (**2003**) theorem to better characterize the mutual information between an agentic AI system's world model and the external world.

- Explored how such extensions might support more empirical approaches to quantifying goal alignment.
- Supervised by MATS Scholar Alex Altair and Dr. Alfred Harwood

Citation Networks & Innovation | *Master's Thesis*

Sumer 2025–Spring 2026

- Developed a new analytical probability distribution for the in-degree of nodes in finite-size DAGs whose growth is governed by cumulative advantage (high in-degree nodes are more likely to attract more in-degrees) and homophily (nodes with similar characteristics being more likely to connect)
- Developed a new Kolmogorov–Smirnov and parametric bootstrap based goodness-of-fit algorithm to account for the non-independent, finite-size, discrete data subsets arising in DAG models of real-world networks
- Supervisor: Associate Professor Tim Evans. Theoretical Physics Group, Centre for Complexity Science
- Supervisor reviewed and confirmed results as extensions to current literature and the paper is in the process of being shared with visiting academics

Hate Speech Detection Benchmarks | *Advanced Humanities Project* Winter 2025–Spring 2026

- Researched bias in benchmark datasets for supervised machine learning hate speech detection algorithms and its implications for equality of opportunity in algorithmic decision-making
- Applied Fishkin's bottleneck theory, adapted by Jain et al. for algorithmic fairness, to analyse benchmarks as gatekeeping mechanisms in access to algorithmically governed spaces. Proposed a policy framework showing that reducing bottleneck severity can mitigate harms associated with both restricted access and benchmark bias
- Supervisor: Dr Catherine Webb: Science Communication Teaching Fellow

BERT Economic Policy Inference Software | *Imperial HPC + BSc Thesis* Summer–Winter 2024

- Developed a Bidirectional Encoder Representations from Transformers (BERT)-based NLP tool to extract policy topics, producing, for example, the per document probability of actionable insight for economic climate policy documents, achieving near-industry accuracy (ClimateBERT dataset)
- Developed into undergraduate thesis on uncertainty quantification in large language models; investigated inverse Fisher Information based variance estimates, implemented Monte Carlo dropout for transformer uncertainty and analysed information loss from UMAP dimensionality reduction
- Worked independently over the summer under guidance from the HPC Research Group, before continuing under the supervision of Dr Aidan Crilly: Wendy Schmidt AI Fellow

Employment

Product Engineer Intern at Mindlace Ltd

Summer 2025

- Led development of an AI agent-driven planning application MVP. Built multi-agent architectures that autonomously coordinated complex booking workflows
- Used modern AI agent software (n8n, Supabase, OpenRouter, Pinceone) to create the entirely unsupervised booking of special requirement meeting itineraries requested by users

Science, English, and Maths Teacher and Tutor

2021 – 2024

- Tailored explanations to meet diverse learner needs, effectively supported a range of abilities and neurodivergence in pupils
- Demonstrated leadership in 4-month weekly placement independently teaching GCSE and A-level Physics to classrooms of 30 students at Brentside High School, Ealing

Landways Management

Summer 2021

- Evidenced reconfiguration of weak network strengths in a football stadium and subsequent maintenance of the improved systems
- Installation of hardware based on technical understanding

Ove Arup Engineering Internship

Spring 2019

- Implemented creative solutions to reduce project timelines on internal software
- Punctual and professional temperament

Software Experience

- Python. Relevant libraries: PyTorch, SciPy, NumPy, Pandas, Matplotlib, Plotly
- NetLogo
- Minimal Viable Product Software Engineering tools: n8n, Shadcn/ui, Supabase, OpenRouter, Pinecone, Firecrawl, Tavily

Hobbies & Interests

Football & Running

- Imperial Men's Football 1st and 2nd teams, previously acting captain of 2nd team
- Enjoy running. Previously competitively at nation level however now recreationally

Journalism & Music

- Comment Piece writer for, Felix, the Imperial College student newspaper. Writing on: Artificial Intelligence
- To wind down, I like to play the guitar